

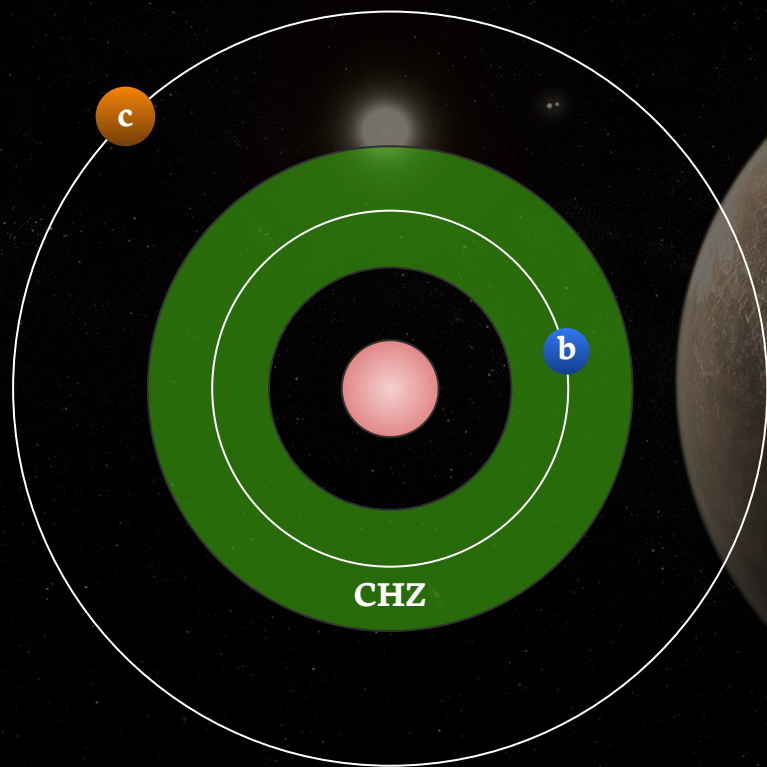
Orbital and Rotational Evolution of Proxima Centauri b



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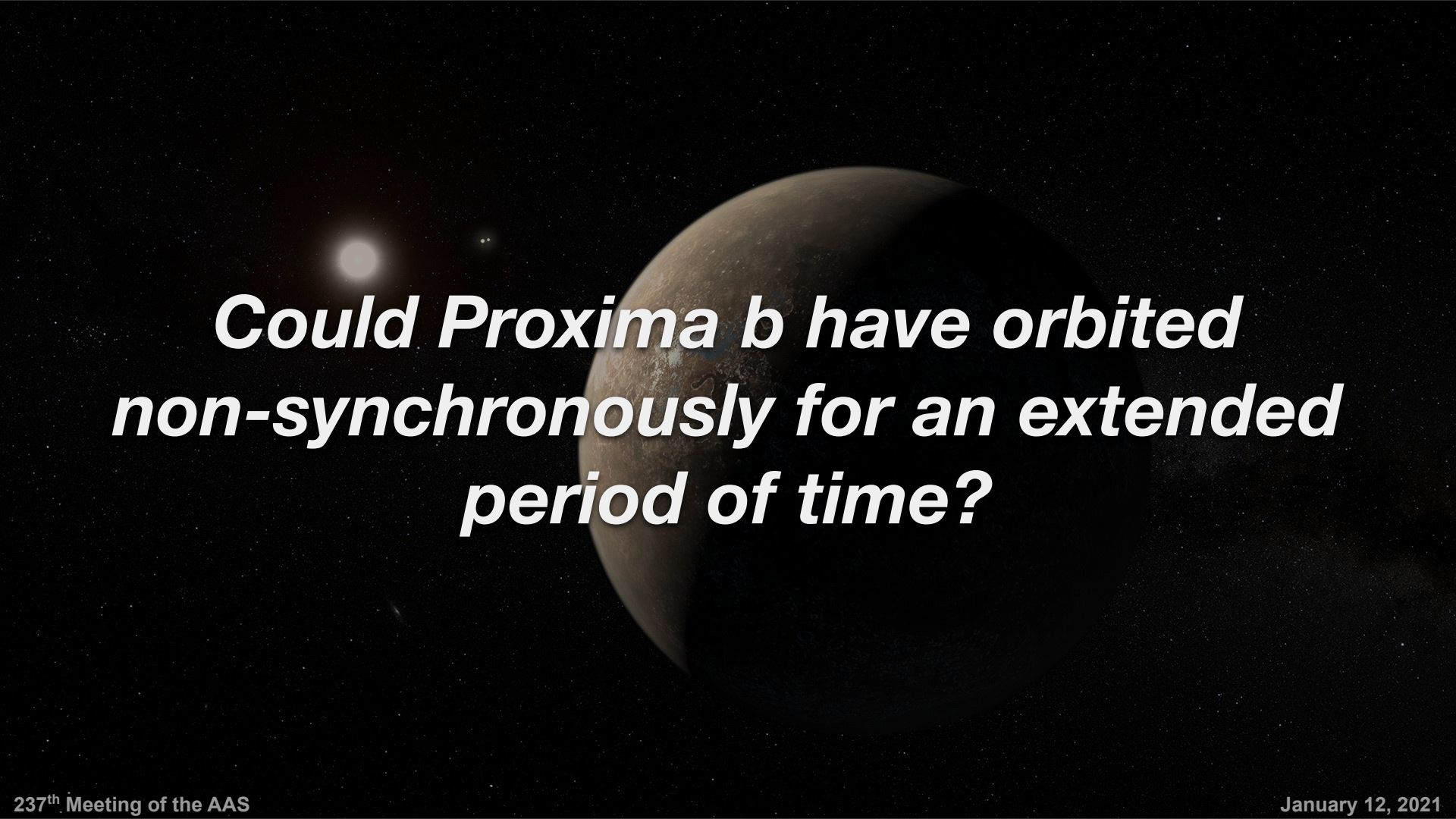
Proxima Centauri planetary system



- Proxima Centauri
 - M-dwarf
 - Closest star to the Sun (4.2 ly)
- Proxima b
 - Identified in 2016
 - ~Earth-mass planet
 - Within CHZ (0.0485 AU)
- Proxima c
 - Identified in 2020
 - Super-Earth
 - Outside of CHZ (1.48 AU)

Habitability of Proxima b

- Tight orbit (semi-major axis 12% of Mercury's)
- Expected to be a **synchronous rotator**
- Leads to uneven heating of the surface—**possible barrier to abiogenesis**
- Capture would occur quickly without second planet
- Our goal is to find how Proxima c affects time of capture



***Could Proxima b have orbited
non-synchronously for an extended
period of time?***

VPLanet

- Modular code for integrating solar system evolution
- We consider three necessary processes:
 - Tidal forces
 - Planet-planet perturbations
 - Stellar evolution (from Baraffe et al. 2015)
- Tidal models
 - “Constant phase lag” (CPL)
 - “Constant time lag” (CTL)



VPLanet Developers Workshop, week of June 7th!

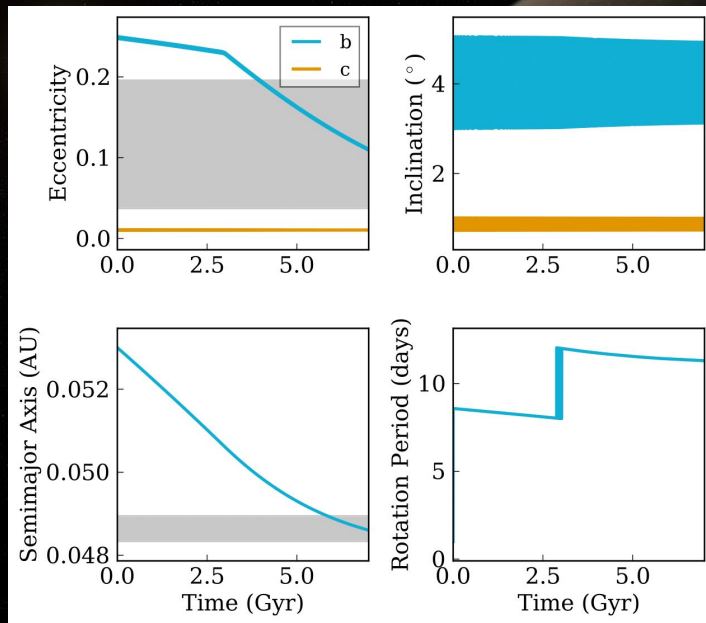
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Initial conditions

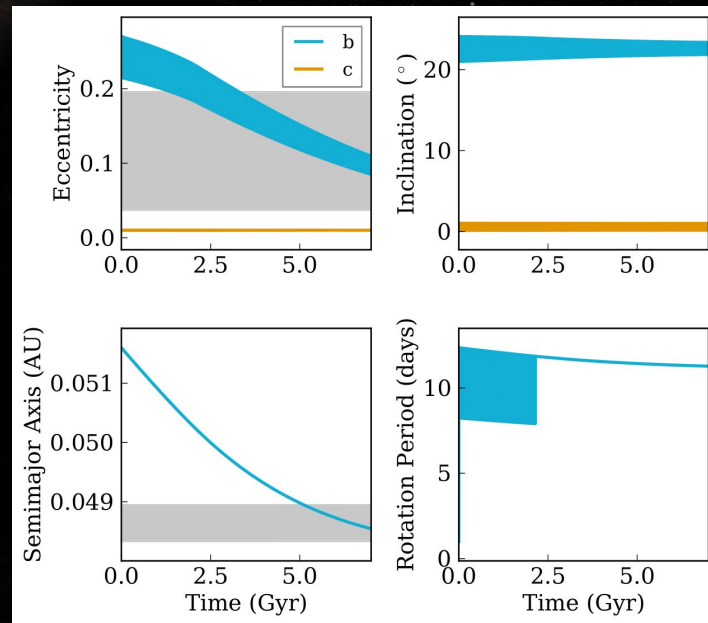
- Most orbital elements of Proxima b are **unconstrained**
- Semi-major axis and eccentricity calibrated with observational values
- Initial rotational period is arbitrary, must be much less than 11 days
- Inclination between planets set to **2°** and **20°**

Evolution under CPL model

2° case



20° case



Grey bars are
ranges for
observational
values

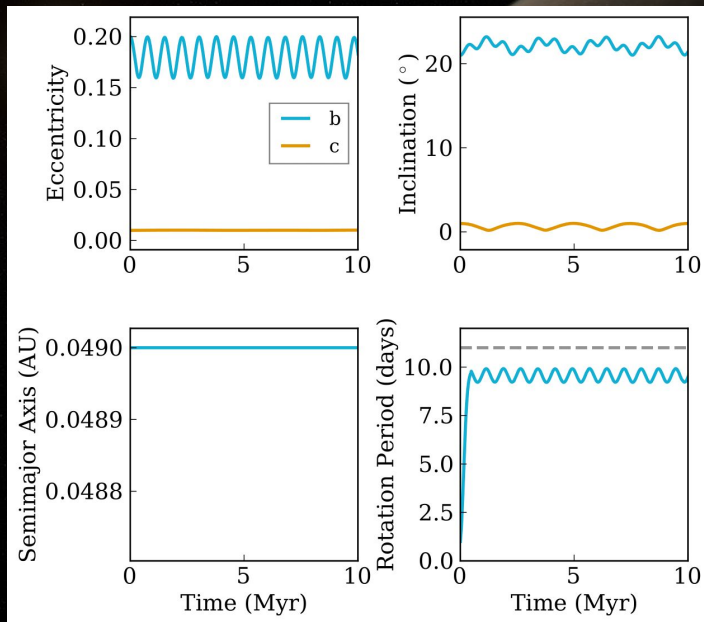
In both evolutions,
**no capture for
nearly 3 Gyr**

Evolution under CTL model

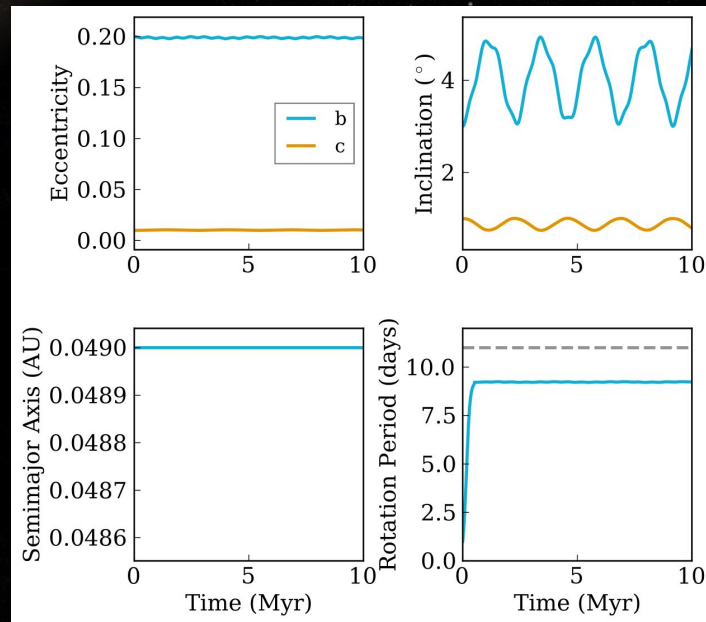
Dashed line shows
synchronous
rotational period

In both evolutions,
planet **captured
into resonance
before
synchronous
rotation**

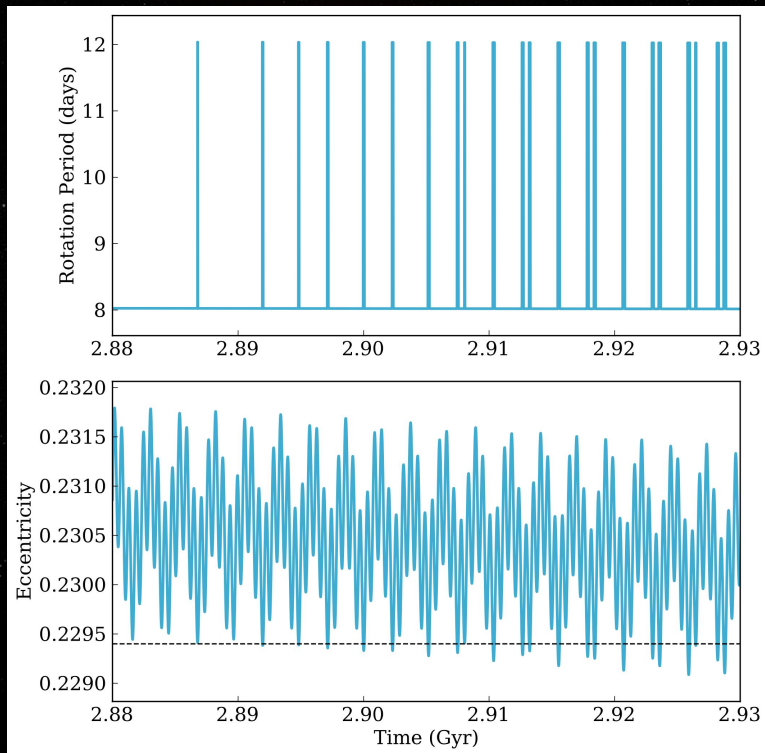
2° case



20° case



Capture into synchronous rotation state



- CPL figures show oscillatory rotational period before capture
- VPLanet **discretizes possible rotational periods** under the CPL model
- Leads to **“jumps” between 3:2 and 1:1 states** where critical eccentricity is crossed
- Not to say that such behavior is physically impossible

Summary

- **Planet-planet perturbations** from Proxima c stymie synchronous capture
 - Under CPL model, prevented **up to 3 Gyr**
 - Under CTL model, prevented **indefinitely**
- If synchronous rotation is prohibitive of life, then we have established **life still could have formed** on Proxima b